

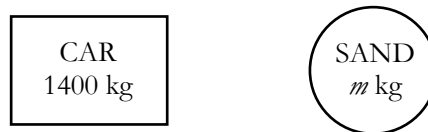
FORCE DIAGRAMS

10/09/25

A Newtonian Model Force Diagram (NMFD) is a schematic representation of the relevant forces on an object or objects and those objects' accelerations. There are some simple rules to follow when drawing an NMFD.

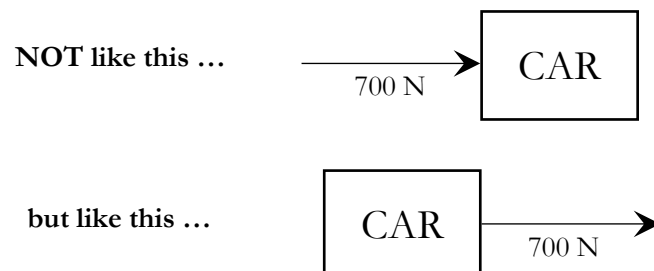
If the object can be modelled as a **particle**, it should be represented by a simple shape like a circle or a rectangle. A description can be written inside the shape to indicate what it represents. If the object's mass is known, this can be written in the shape, or a variable can be used.

For instance:



The forces on an object should be represented as **single-headed** arrows pointing in the direction in which they act and **emanating from the object**. An arrow might have one of two labels: a description of the force (e.g. 'force of table on book'), or the size of the force, using a variable if unknown (e.g. R Newtons).

This clause in bold is important. In an NMFD of several objects it is important to sort out which forces are acting on which objects without confusion. So ...



The acceleration of an object should be represented as a **doubled-headed** arrow **set apart** from an object, pointing in the direction of acceleration. If the acceleration is zero, the object is said to be in **equilibrium**.

For instance:

